Marwan M. Al-Falah

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RESEARCH PROFILE

MSc researcher in Intelligent Systems with research experience across artificial intelligence, machine learning, multimodal systems, agentic architectures, NLP, and research software engineering. Recent work includes governance-aware intelligent systems, browser-grounded multimodal agents, clinical AI, and audio-visual process monitoring. I am interested in PhD research across AI and intelligent systems, including adjacent directions where rigorous experimentation, reliable system design, and meaningful real-world impact are central.

Research Interests: Artificial Intelligence • Machine Learning • Intelligent Systems • Multi-Agent & Agentic Systems • Multimodal AI • NLP • Trustworthy & Evaluated AI • Autonomous Systems • Applied AI

EDUCATION

Peter the Great St. Petersburg Polytechnic University (SPbPU), Russia Sep 2025 - Aug 2027 (expected)

MSc in Intelligent Systems | Expected graduation: Aug 2027

- Thesis: Experimental Study of the Advantages and Disadvantages of Centralized and Decentralized Multi-Agent System Architectures Based on LLMs.

Knowledge & Modern Sciences University (KMSU), Yemen Sep 2019 - Jul 2023

BSc in Information Technology | High Honors | **GPA: 3.71/4.0**

- Thesis: BotSappy - NLP-Based Customer Service Chatbot Platform.

RESEARCH EXPERIENCE & CURRENT COLLABORATIONS

ITMO Operating-Room Multimodal Monitoring Research Collaboration Current

- Collaborating with an ITMO University research group on intelligent monitoring of medical and technological processes in the operating room using video and speech analytics; the ongoing work is targeting a Q1 journal submission.
- Contributing to the visual-analysis component, including observable activities, temporal relationships, protocol-deviation analysis, and cases that require expert review.

SMILES 2026 Research Project - WebPilot | Nanjing University × Skoltech, Suzhou, China Jul 2026

- Selected for on-site participation in SMILES 2026 and contributed to WebPilot: Browser-Grounded Multimodal Web Coding Agent; co-author of the manuscript submission.
- Compared single-pass, fixed-pipeline, and autonomous agent approaches using browser feedback across text-, image-, and video-guided web generation tasks.

Vendor-Agnostic Industrial Orchestrator | Research Prototype Current

- Developed a deterministic, governance-aware prototype separating planning, capability matching, policy checks, and execution; evaluated behavior across 240 scenarios with execution traces, approval and recovery experiments; first author of the associated manuscript.

Deep Synaptica - Agent Observability & Evaluation Platform 2026

- Developing an open-source platform for recording agent execution traces and inspecting runs through an API and dashboard, with emphasis on measurable evaluation and reproducible analysis.

Clinical Documentation System | Research Prototype 2026

- Built an NLP pipeline that converts clinical speech into structured SOAP notes with explicit workflow checks and a human-review path.

TeraFinder / Multi-Agent Search Filtering | ITMO × Sber Open Source BootCamp 2025

- Contributed to a LangGraph-based multi-source retrieval and synthesis system for evidence-oriented search in a collaborative open-source research setting.

SELECTED PEER-REVIEWED PUBLICATIONS

1. Yousif A. Alhaj, **Marwan M. Al-Falah**, et al. "Beyond URL Analysis: Evaluating Content-Based Phishing Detection Using HTML Structural Features." *Artificial Intelligence Topics and Applications*, 2(1), 80-97, 2026. **Corresponding author.**
2. E. S. Gebel, **Marwan M. Al-Falah**, and Yousif A. Alhaj. "Designing a Multi-Agent Intelligent Control System for Real-Time Clinical Documentation with Safety Oversight." *XXIII Youth and Modern Information Technologies, TPU*, 2026.
3. Yousif A. Al-Haj, **Marwan M. Al-Falah**, et al. "Smart Diagnosis System for Diabetic Retinopathy." *CAISAIS, IEEE*, 2023.
4. Yousif A. Al Haj, **Marwan M. Al-Falah**, et al. "An Efficient Machine Learning Algorithm for Breast Cancer Prediction." *CAISA, Springer Nature*, 2023.

RESEARCH MANUSCRIPTS & ONGOING WORK

Marwan M. Al-Falah, Elena S. Gebel, and Naif M. Al-Falah. "A Deterministic Governance-Aware Agentic Framework for Vendor-Agnostic Industrial Workflow Orchestration." *Research manuscript; first author.*

Yusong Dong, Nuojing Liang, Junchen Yu, **Marwan M. Al-Falah**, Xinping Lei, and Jiaheng Liu. "WebPilot: Browser-Grounded Multimodal Web Coding Agent." *SSPP 2026 manuscript submission via OpenReview.*

Intelligent Monitoring of Medical and Technological Processes in the Operating Room Based on Video and Speech Analytics. Ongoing collaborative research with an ITMO University group; **targeting a Q1 journal submission.**

TECHNICAL SKILLS

AI & Research: Artificial Intelligence, Machine Learning, Intelligent Systems, Multi-Agent & LLM Agents, Multimodal AI, NLP, Video Analytics, Supervised Learning, Trustworthy AI, Model Interpretability, Experimental Evaluation

Frameworks & Engineering: Python, LangGraph, FastAPI, REST APIs, Docker, Git, Testing, Research Prototyping

Data & Analysis: SQL/MySQL, R, Tableau, scikit-learn, SHAP

Web: JavaScript, PHP, ReactJS

PROFESSIONAL EXPERIENCE

Fullstack Web Developer (Remote) - Nexla for Software and Creative Solutions Jul 2024 - Aug 2025

- Developed Laravel backend services and JavaScript interfaces, integrated APIs, reviewed code, optimized performance, and resolved production issues.

Data Analyst - Human Group Organization for Relief & Development (NGO) Oct 2019 - Jul 2024

- Prepared analyses and reports and coordinated information between management and field teams.

SELECTED TRAINING & LEADERSHIP

Generative & Agentic AI - Coimbatore Institute of Technology (CIT), India Jan 2026

Training in building and orchestrating autonomous AI systems for real-world applications.

Information Technology Enhancement Program (ITEP) - DJIV, Berlin, Germany Feb-Apr 2024

Collaborative development, software architecture, engineering practices, and software operations.

Founder & Lead - Google Developer Student Clubs (GDSC) Jul 2022 - Jul 2023

Led a university developer community and student technical activities.

LANGUAGES

Arabic (Native) • **English** (Fluent) • **Russian** (Intermediate) • **French** (Basic)